Lecture 7 Logistic Regression I

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

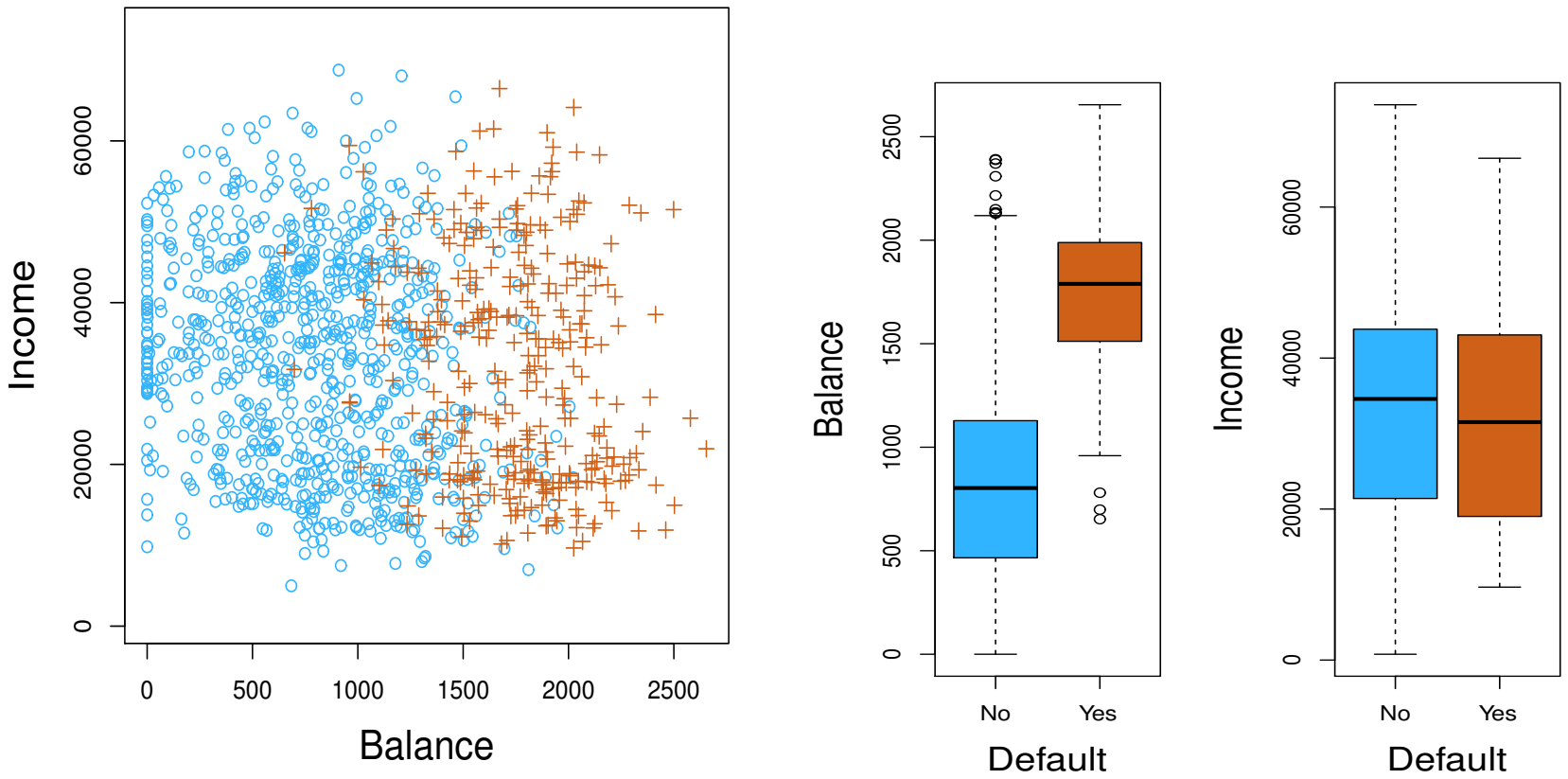
Qualitative Response

- There are many qualitative response taking values in an unordered set $\mathcal{C}$ such as
  `eye color ∈ {brown; blue; green}`.
- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $C(X)$ (learn a rule) that takes as input the feature vector X and predicts its value for Y; i.e. $C(X) \in \mathcal{C}$.
- Often we are more interested in estimating the probabilities that X belongs to each category in $\mathcal{C}$.
- For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a classification fraudulent or not.

Examples

- A person arrives at the emergency room with a set of symptoms that could possibly be attributed to one of three medical conditions:
  - Which of the three conditions does the individual have?
- An online banking system manages transactions, storing user's IP address, past transaction history, and so forth
  - Is the transaction fraudulent or not?
- A biologist collects DNA sequence data for a number of patients with and without a given disease
  - Which DNA mutations are deleterious (disease-causing) and which are not?

Credit Card Default



Individuals who defaulted in a given month in orange, and did not in blue.

Linear Regression Model

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- Suppose for the Default classification task that we code

$$Y = \begin{cases} 1 & \text{if Yes} \\ 0 & \text{if No} \end{cases}.$$

  Can we simply perform a linear regression of Y on X and classify as Yes if $\hat{Y} > 0.5$?

- In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to linear discriminant analysis which we discuss later.
- Since in the population $\mathrm{E}(Y|X = x) = \Pr(Y = 1|X = x)$, we might think that regression is perfect for this task.
- However, linear regression might produce probabilities less than zero or bigger than one. Logistic regression is more appropriate.

Why not Linear Regression?

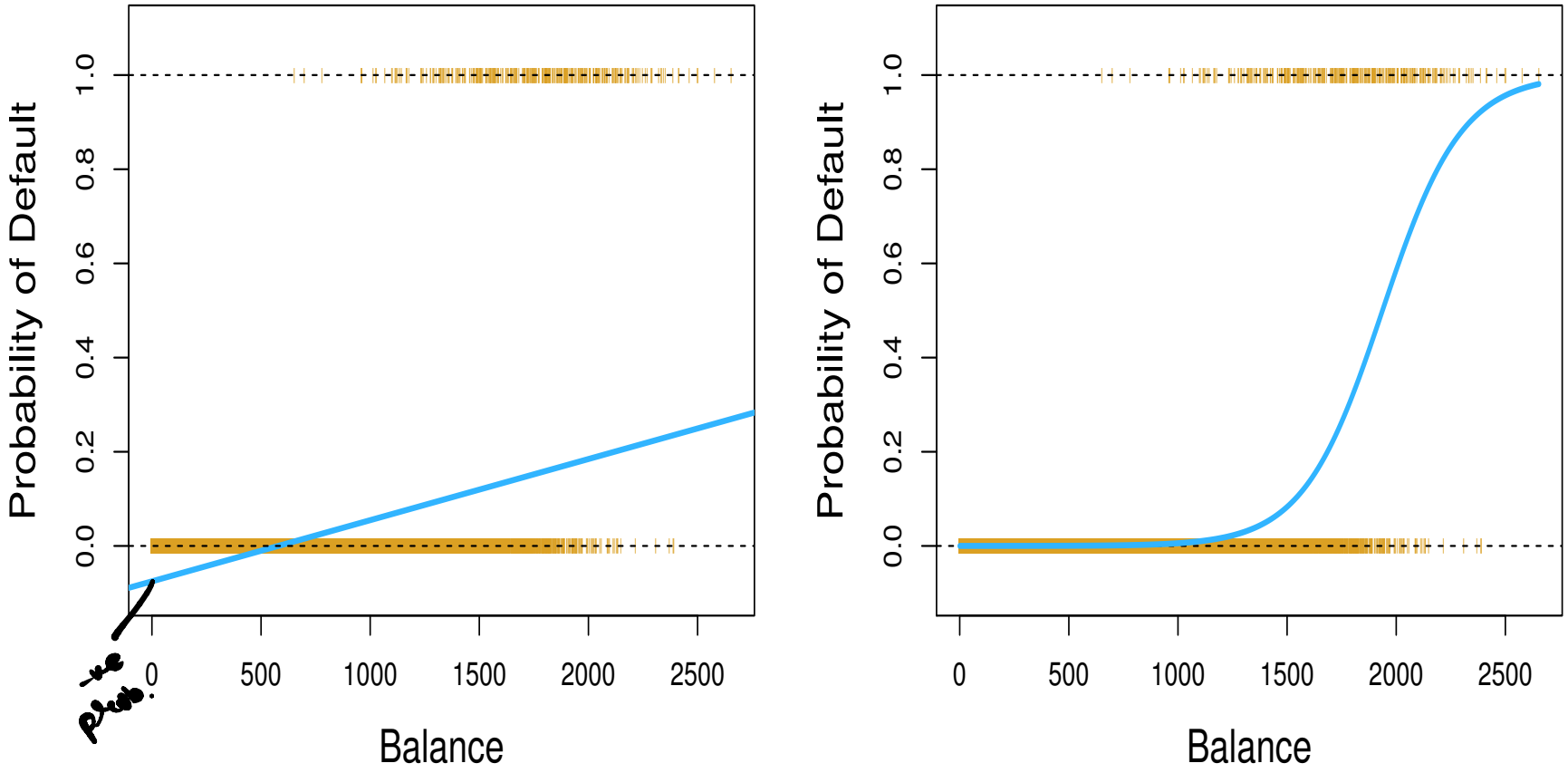
- Suppose that we are trying to estimate the medical condition of a patient in the emergency room based on her symptoms
- There are three possibilities: stroke, drug overdose and epileptic seizure
- We could consider encoding these values as a quantitative response variable, y, as

$$y = \begin{cases} 1 & \text{if} \quad \text{stroke} \\ 2 & \text{if} \quad \text{drug overdose} \\ 3 & \text{if} \quad \text{epileptic seizure} \end{cases}$$

- However, we are implicitly saying that the «difference» between drug overdose and stroke is the same as the «difference» between epileptic seizure and drug overdose, which does not make much sense

Credit data example

Linear Reg. behaves like this



The orange marks indicate the response Y, either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

Logistic Regression

- Logistic regression (Cox, 1958) is probably the most fundamental and widely used statistical tool to model binary or multi-category response variable.
- Denote $p(X) = \Pr(Y = 1|X)$ consider using balance to predict default. Logistic regression uses the form

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}.$$

Cont and diff,
Bw 0 and 1,
always positive
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- It is easy to see that no matter what values β_0, β_1 or X take, $p(X)$ will have values between 0 and 1.
- A bit of rearrangement gives

P(y=1|x)

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X.$$

Linear model

P(y = 0|x)

- This monotone transformation is called the log odds or logit transformation of $p(X)$.

Estimation

- We use maximum likelihood to estimate the parameters.

To find the parameters, this is for every point(different xi's)

$$l(\beta_0, \beta_1) = \Pi_{i:y_i=1} p(x_i) \Pi_{i:y_i=0} (1 - p(x_i)).$$

- This likelihood gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.
- Most statistical packages can fit linear logistic regression models by maximum likelihood. In R we use the glm function.

```
> glm.fit=glm(default~balance,data=defaultData,family=binomial)
> summary(glm.fit)
Coefficients:
Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.065e+01  3.612e-01  -29.49   <2e-16 ***
balance      5.499e-03  2.204e-04   24.95   <2e-16 ***
---
Signif. codes:  0 ?***? 0.001 ?**? 0.01 ?*? 0.05 ?.? 0.1 ? ? 1
```

Interpretation

- Interpreting what β_1 means is not very easy with logistic regression, simply because we are predicting $\Pr(Y = 1|X)$ and not Y.
- If $\beta_1 = 0$, this means that there is no relationship between Y and X.
- If $\beta_1 > 0$, this means that when X gets larger so does the probability that $Y = 1$.
- If $\beta_1 < 0$, this means that when X gets larger, the probability that $Y = 1$ gets smaller.
- But how much bigger or smaller depends on where we are on the slope.

Hypothesis Testing

- We still want to perform a hypothesis test to see whether we can be sure that are β_0 and β_1 significantly different from zero.
- We use a z test instead of a t test, but of course that doesn't change the way we interpret the p-value
- Here the p-value for balance is very small, and β_1 is positive, so we are sure that if the balance increase, then the probability of default will increase as well.

```
Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.065e+01  3.612e-01  -29.49   <2e-16 ***
balance      5.499e-03  2.204e-04   24.95   <2e-16 ***
---
Signif. codes:  0 ?***? 0.001 ?**? 0.01 ?*? 0.05 ?.? 0.1 ? ? 1
```

Prediction

- What is our estimated probability of `default` for someone with a balance of 1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.5613+0.0055\times 1000}}{1+e^{-10.5613+0.0055\times 1000}} = 0.006.$$

- The predicted probability of default for an individual with a balance of \$1000 is less than 1%.
- For a balance of \$2000, the probability is much higher, and equals to 0.586(58.6%).

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.5613+0.0055\times 2000}}{1+e^{-10.5613+0.0055\times 2000}} = 0.586.$$

```
> predict(glm.fit, list(balance = c(1000,2000)),type="response")
1           2
0.005752145 0.585769370
```

Summary and Remark

- Introduction
- Logistic regression and estimation
- Hypothesis testing and prediction
- Read textbook Chapter 4 and `Python` code